Simulating Photoinduced Electron Dynamics for X-ray Absorption Spectroscopy and X-ray Natural Circular Dichroism in Molecules

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Introduction

Chirality is central to chemistry and biochemistry and drug design [1]. X-ray sources now extend chiroptical spectroscopy to core excitations, where resonant X-ray absorption (XAS) combines site specificity with strongly enhanced circular dichroism (XNCD) [2]. We address the need for theoretical assignments by propagating the time-dependent Schrödinger equation under an explicit pulse to compute XAS and XNCD in molecules with WaveT [3], enabling interpretation of X-ray chiroptical spectra and ground- and valence excited-state pump–probe simulations [4].

Methods

Real-time propagation (WaveT): XAS and XNCD are obtained from the time-dependent Schrödinger equation in the length gauge or the velocity gauge [5]:

$$i\frac{\mathrm{d}}{\mathrm{d}t}|\Psi(t)\rangle = \left(\hat{H}_0 + \hat{H}_I(t)\right)|\Psi(t)\rangle\,, \tag{1}$$

where $\hat{H}_I(t)$ is the interaction Hamiltonian for the length gauge ($\hat{H}_I(t) = -\hat{\vec{\mu}}\cdot\vec{F}(t)$) or the velocity gauge ($\hat{H}_I(t) = \hat{\vec{p}}\cdot\vec{A}(t)$), and driven by a Gaussian-enveloped oscillatory pulse $\vec{F}(t) = \vec{F}_m \exp\left(-\frac{(t-t_0)^2}{2\sigma^2}\right)\sin(\omega_c t) = -\frac{\partial\vec{A}(t)}{\partial t}$, with carrier frequency ω_c, center t_0, width σ, and peak amplitude $\vec{F}_m$.

State-space expansion: $|\Psi(t)\rangle$ is expanded in TDDFT [6] eigenstates of $\hat{H}_0$. For ground-state (GS) spectra (core–valence separation, CVS):

$$|\Psi(t)\rangle = C_0(t)\,|0\rangle + \sum_{\lambda_c=1}^{N_c} C_{\lambda_c}(t)\,|\lambda_c\rangle\,. \tag{2}$$

For excited-state (ES) spectra:

$$|\Psi(t)\rangle = C_0(t)\,|0\rangle + \sum_{\lambda_c=1}^{N_c} C_{\lambda_c}(t)\,|\lambda_c\rangle + \sum_{\lambda_v=1}^{N_v} C_{\lambda_v}(t)\,|\lambda_v\rangle\,. \tag{3}$$

Spectra (linear response): With $\Delta\mu_q(t) = \mu_q(t) - \mu_q(0)$, $\Delta m_q(t) = m_q(t) - m_q(0)$ the time-dependent induced dipole moment and magnetic moment, along direction q, and $F_n^0(\omega)$ the Fourier transform of the field along direction n, the isotropic XAS and XNCD spectra are

$$P^{\mathrm{XAS}}(\omega) = \frac{1}{3}\sum_{n\in\{x,y,z\}} P^{\mathrm{XAS}}_{nn}(\omega), \quad P^{\mathrm{XNCD}}(\omega) = \frac{1}{3}\sum_{n\in\{x,y,z\}} P^{\mathrm{XNCD}}_{nn}(\omega), \tag{4}$$

$$P^{\mathrm{XAS}}_{nq}(\omega) = \frac{1}{2\pi F_n^0(\omega)}\int_0^\infty \Delta\mu_q(t)\, e^{i(\omega+i\Gamma)t}\,\mathrm{d}t,$$
$$P^{\mathrm{XNCD}}_{nq}(\omega) = \frac{i}{2\pi\omega F_n^0(\omega)}\int_0^\infty \Delta m_q(t)\, e^{i(\omega+i\Gamma)t}\,\mathrm{d}t, \tag{5}$$

where Γ is a damping parameter (average excited-state lifetime).

Simulation pipeline:

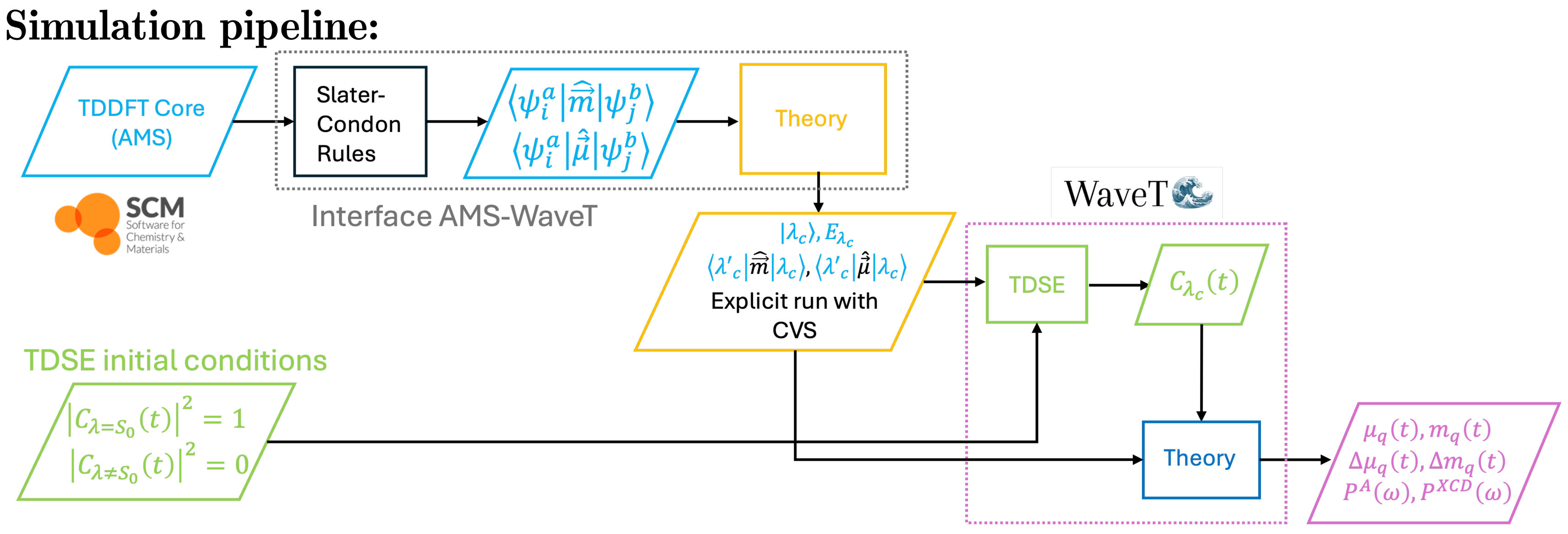

Figure 1: Pipeline for the computation of ground-state XAS and XNCD in molecules.

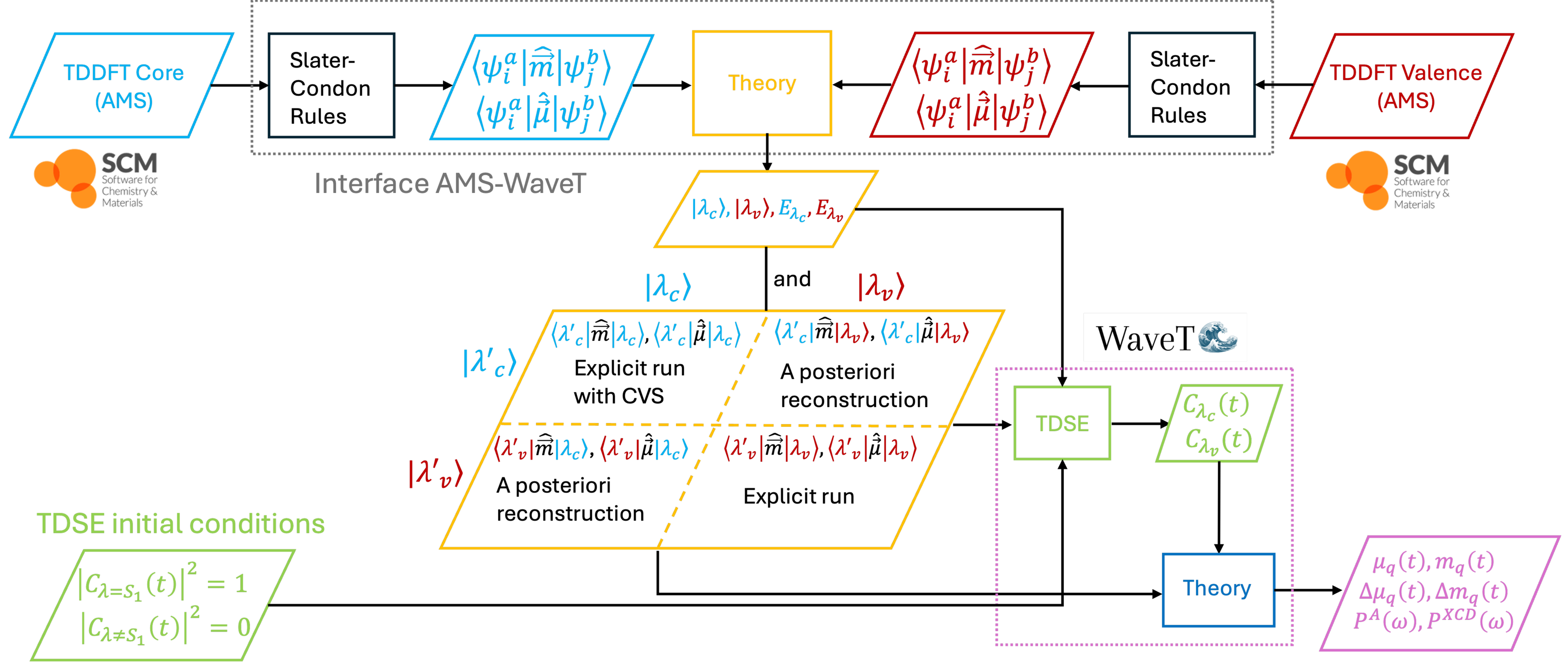

Figure 2: Pipeline for the computation of excited-state XAS and XNCD in molecules.

Acknowledgements

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Computational Details

System and theory. (R)-methyloxirane (geometry from Andersen *et al.* [7], comparison reference); ground-state TDDFT with ADF/AMS, CAM-B3LYP.

Basis sets (BS1–BS4), benchmarked at C K-edge and valence: BS1: C,O augmented TZP, H TZP; BS2: C QZP+1 diffuse, O/H TZP; BS3: C,O QZP+1 diffuse, H DZP; BS4: C,O QZP+1 diffuse, H TZP.

States and propagation. Core TDDFT (CVS, 60 states); valence TDDFT (no orbital freezing, 60 states). C K-edge dynamics: 17 core + 60 valence states; core manifold below the C 1*s* ionization threshold. 97 fs trajectories, $\delta t = 0.12$ as, $I = 10^4\ \mathrm{W\,cm^{-2}}$, $\omega_c = 278$ eV, pulse FWHM 1 fs, $\tau = 1/\Gamma = 200$ a.u.

Results and Discussion

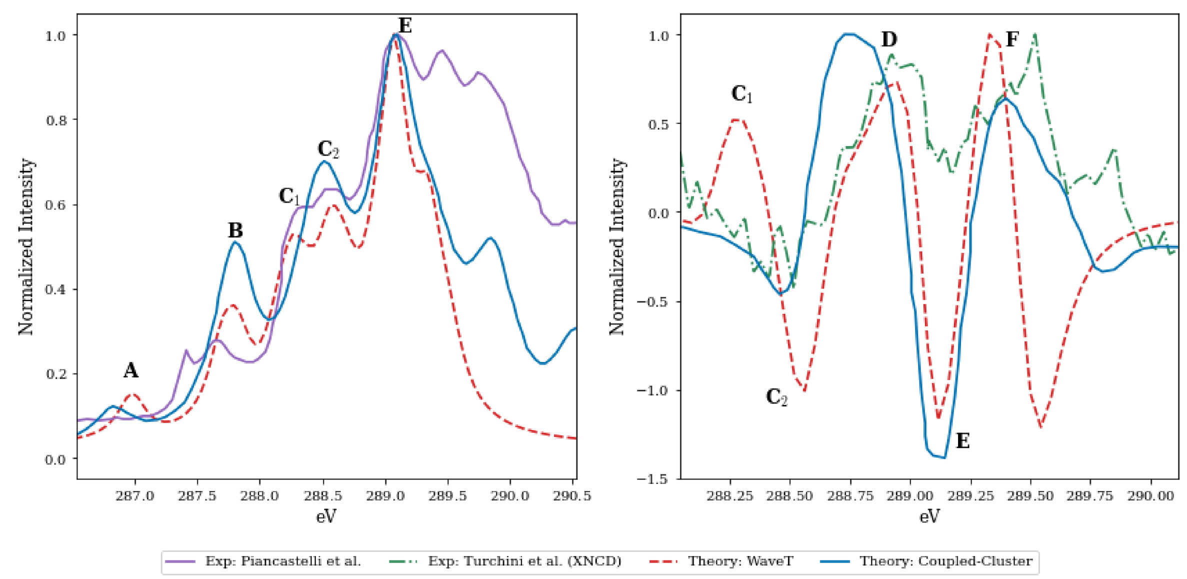

Figure 3: Experimental vs. computed ground-state XAS and XNCD for **C K-edge** of (R)-methyloxirane. Coupled-Cluster data was obtained from Andersen *et al.* [7], while experimental data was obtained from [8, 9]. A rigid shift of 10.6 eV was applied to WaveT results to align with experiment.

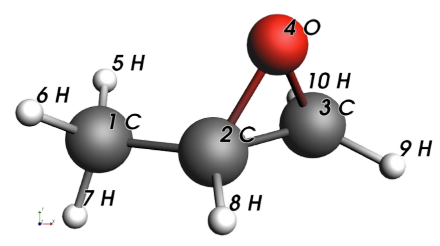

Figure 5: Geometry of (R)-methyloxirane from Andersen *et al.* [7].

Peak	Composition
A	71%: 1S (1C) → LUMO 13%: 1S (1C) → LUMO
B	34%: 1S (1C) → LUMO+3 30%: 1S (1C) → LUMO+1
C_1	26%: 1S (3C) → LUMO+3 25%: 1S (3C) → LUMO
C_2	35%: 1S (3C) → LUMO 19%: 1S (3C) → LUMO+1
D	35%: 1S (3C) → LUMO 19%: 1S (3C) → LUMO+1
E	25%: 1S (2C) → LUMO+14 18%: 1S (2C) → LUMO+3
F	26%: 1S (2C) → LUMO+3 25%: 1S (2C) → LUMO+2

Table 1: Peak composition for ground-state XAS and XNCD.

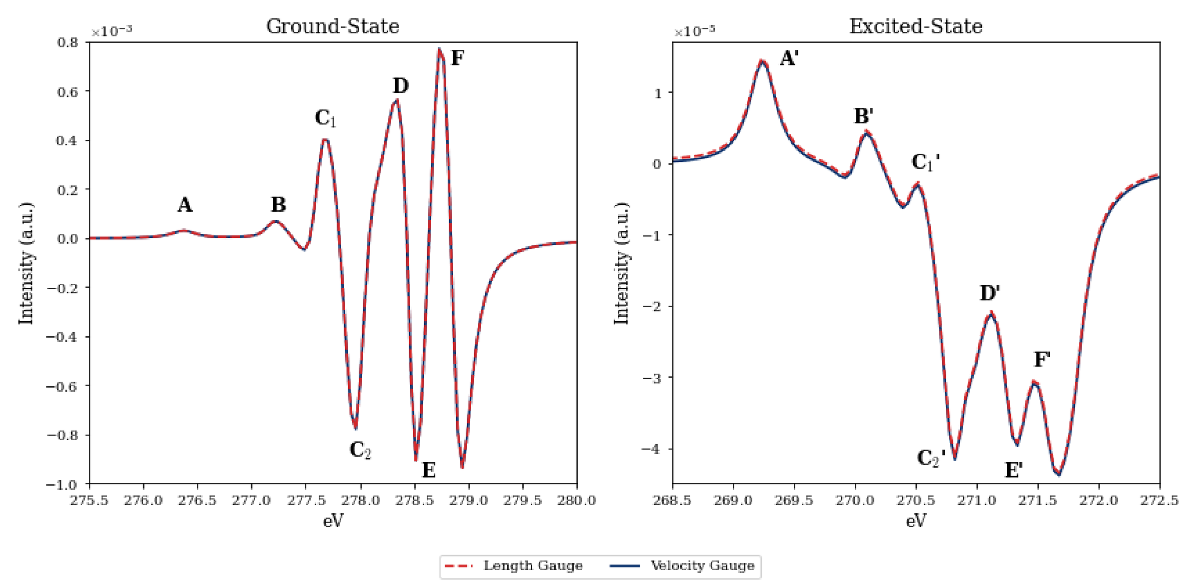

Figure 4: Computed first-excited-state XNCD for (R)-methyloxirane.

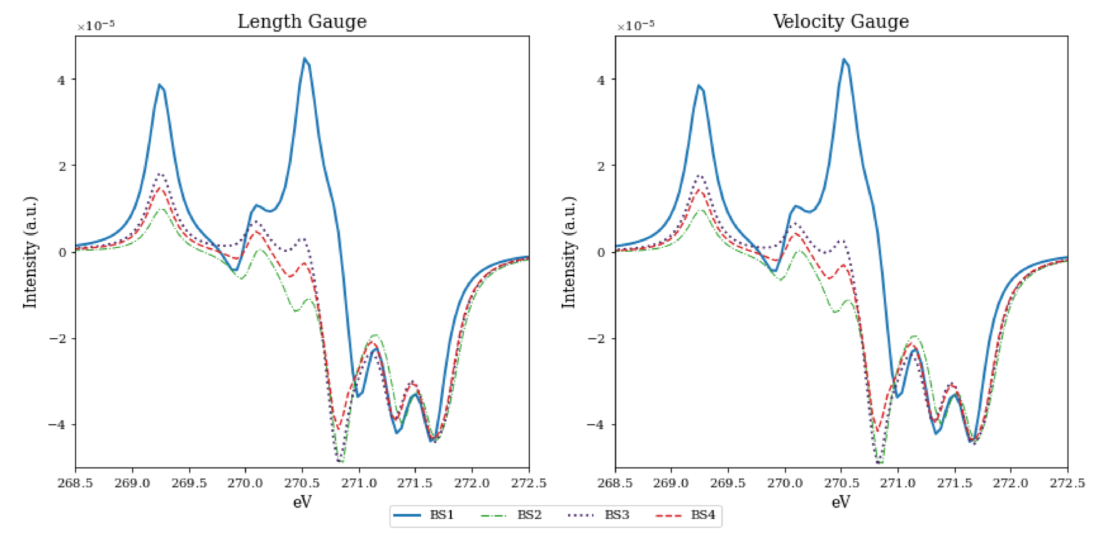

Figure 6: Basis set convergence of computed first-excited-state XNCD for carbon K-edge of (R)-methyloxirane. Core and valence calculations used to obtain the excited-state XNCD share the same basis set.

Conclusions

- WaveT is a general framework for molecular XAS and XNCD, including ES spectra.
- For (R)-methyloxirane, WaveT ground-state XAS and XNCD reproduce experiment and agree with coupled-cluster benchmarks.
- Ground- and excited-state core transitions are assigned unambiguously.
- Ongoing work for including solvent effects (PMM [10] and COSMO).

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